Memory Page Simulator

Visualize FIFO, LRU & OPT Algorithms

Reference Sequence

1 2 3 2 5 6 2 4 1 5

Frame Capacity

3

Run Simulation

FIFO

9

PAGE FAULTS

LRU

8

PAGE FAULTS

OPT

7

PAGE FAULTS

How the Algorithms Work

In all algorithms, if a page fault occurs, we first check if the frame has any free spaces. If so, we simply append the new page into the frame. If there are no free spaces, we do different methods according to different algorithms.

First In First Out (FIFO)

For the FIFO algorithm, we remove pages in the order they entered the frame, similar to a queue. When a new page is needed, it replaces the "oldest" page (front of the queue). After replacement, the next page becomes the new oldest.

Least Recently Used (LRU)

LRU tracks page usage. When a page is referenced, it's moved to the "most recent" position. During replacement, the least recently used page (furthest from recent access) is replaced. This maintains a usage order rather than insertion order.

Optimal Page Replacement (OPT)

This theoretical approach replaces the page that won't be used for the longest time in the future. While it offers the lowest page-fault rate, it's impractical as it requires future knowledge of page requests.